



VELTRIX INTELLIGENCE PLATFORM

VIP COMPLETE SYSTEM WORKFLOW

Enterprise business process, SaaS architecture, data intelligence, AI, automation and operations workflow package

Diagram version 1.0
Generation date 2026-07-21
Repository commit 33a8356da18b5d74831f2a79b727a73779900f50
Source of truth: repository documentation, routes, services, contracts and tests

Status principle: frontend mock-mode behavior is not represented as production backend capability.

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The 16 required workflow diagrams plus package navigation.

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Front matter: Cover - Contents - Diagram legend and conventions
Back matter: Final findings and gaps

Diagram Legend and Conventions

Enterprise visual language and repository-evidence status classification used on every workflow page.

Implemented
Solid blue border: working frontend behavior evidenced by code/tests.

Planned
Dashed grey border: target architecture or documented backend dependency.

Validated successful outcome
Green border: terminal success outcome.

Partially implemented
Orange border: UI/service seam exists but live integration or depth is incomplete.

Missing / blocked
Red border: required production capability has no implementation evidence.

Requires clarification
Dashed violet border: ownership or policy decision is unresolved.

Process / service

Screen / module

Decision

Database

Generated file

External service

Async queue

Worker

Failure path

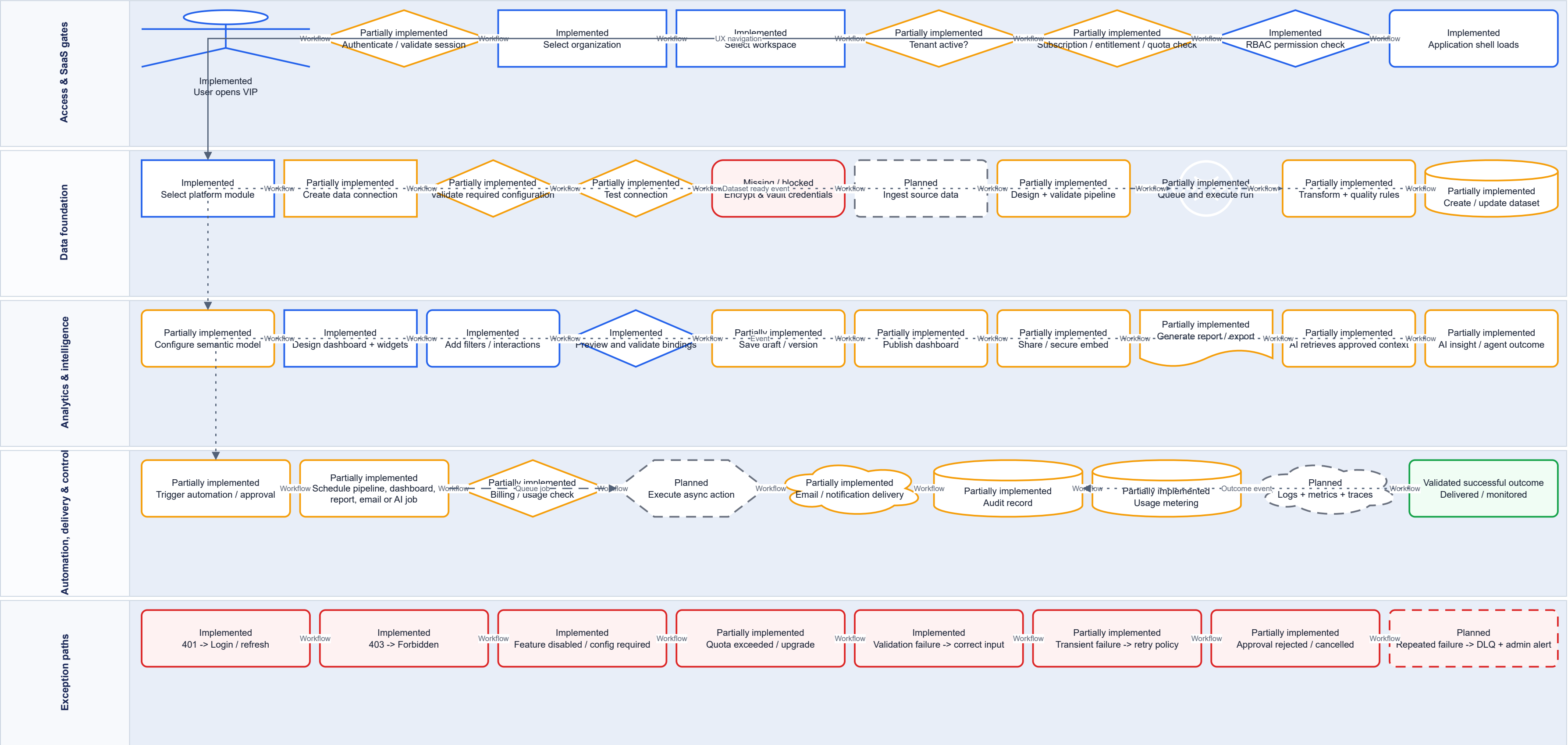
CONNECTORS

Solid arrow = synchronous call / UX step
Dashed arrow = asynchronous job
Dotted arrow = event publication
Red arrow = failure flow
Green arrow = successful completion
Double-headed arrow = true bidirectional communication only
Every key connector carries a protocol or business meaning label.

Status is based on commit 33a8356da18b5d74831f2a79b727a73779900f50. The codebase defaults to mock mode; live backend adapters are integration seams, not proof that target services exist.

Page 1 - VIP End-to-End Platform Workflow

Complete operating model from entry through governed delivery and telemetry.



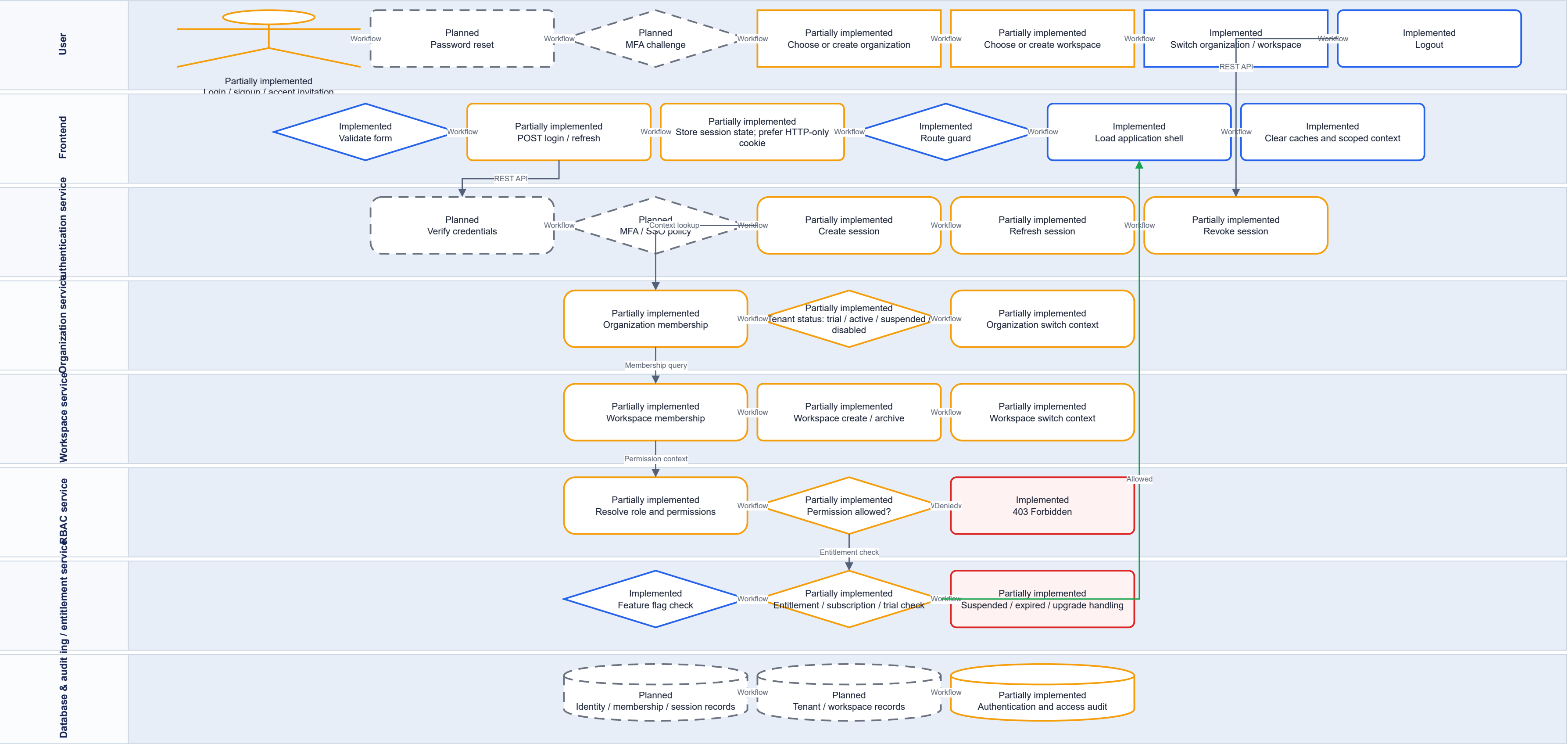
STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 2 - User Access, Authentication and SaaS Workflow

Identity, tenant context, feature access and session lifecycle.



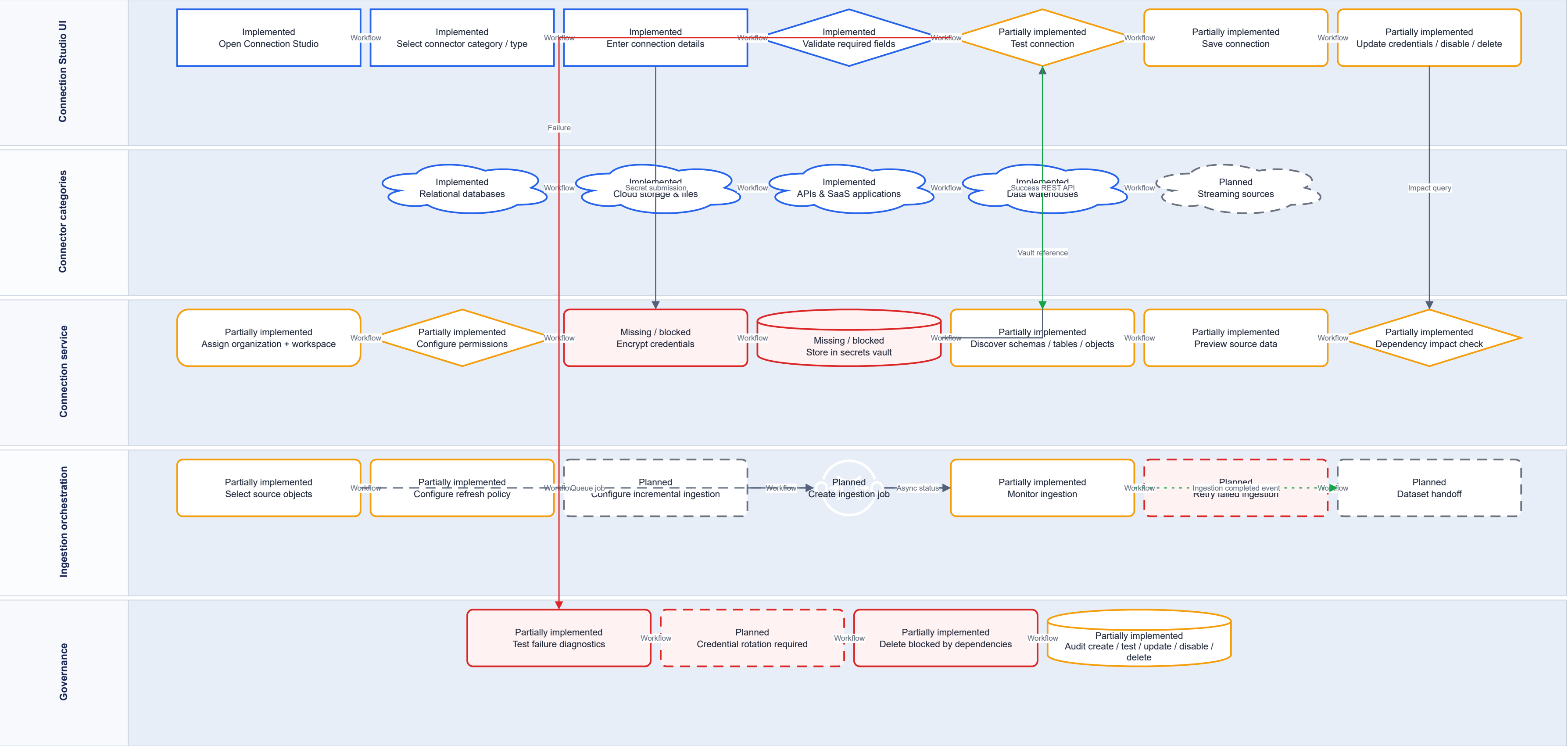
STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 3 - Connection Studio Workflow

Connector onboarding, discovery, ingestion lifecycle and dependency-safe retirement.



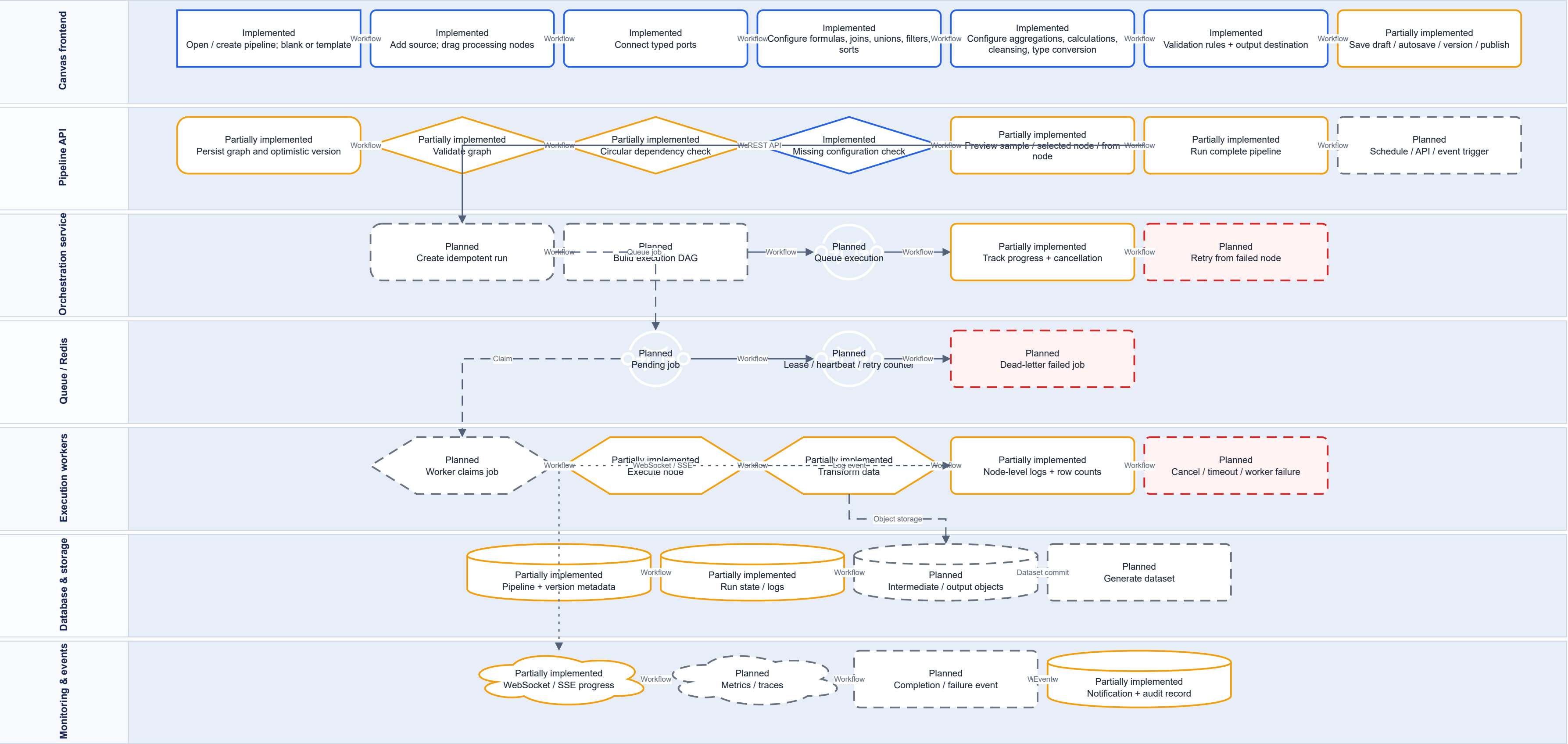
STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 4 - Pipeline Studio Workflow

Alteryx-style visual authoring and planned distributed pipeline execution.

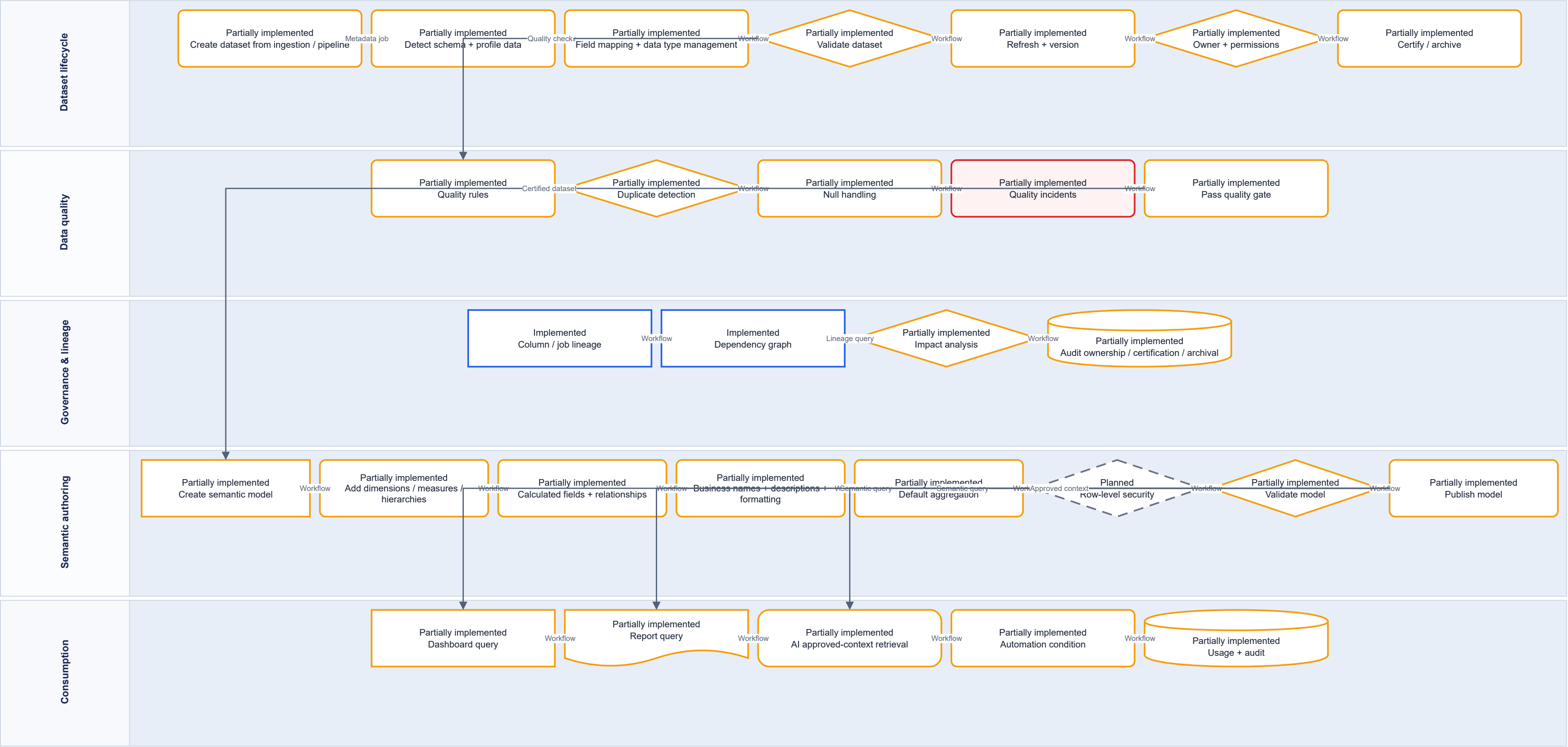


STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

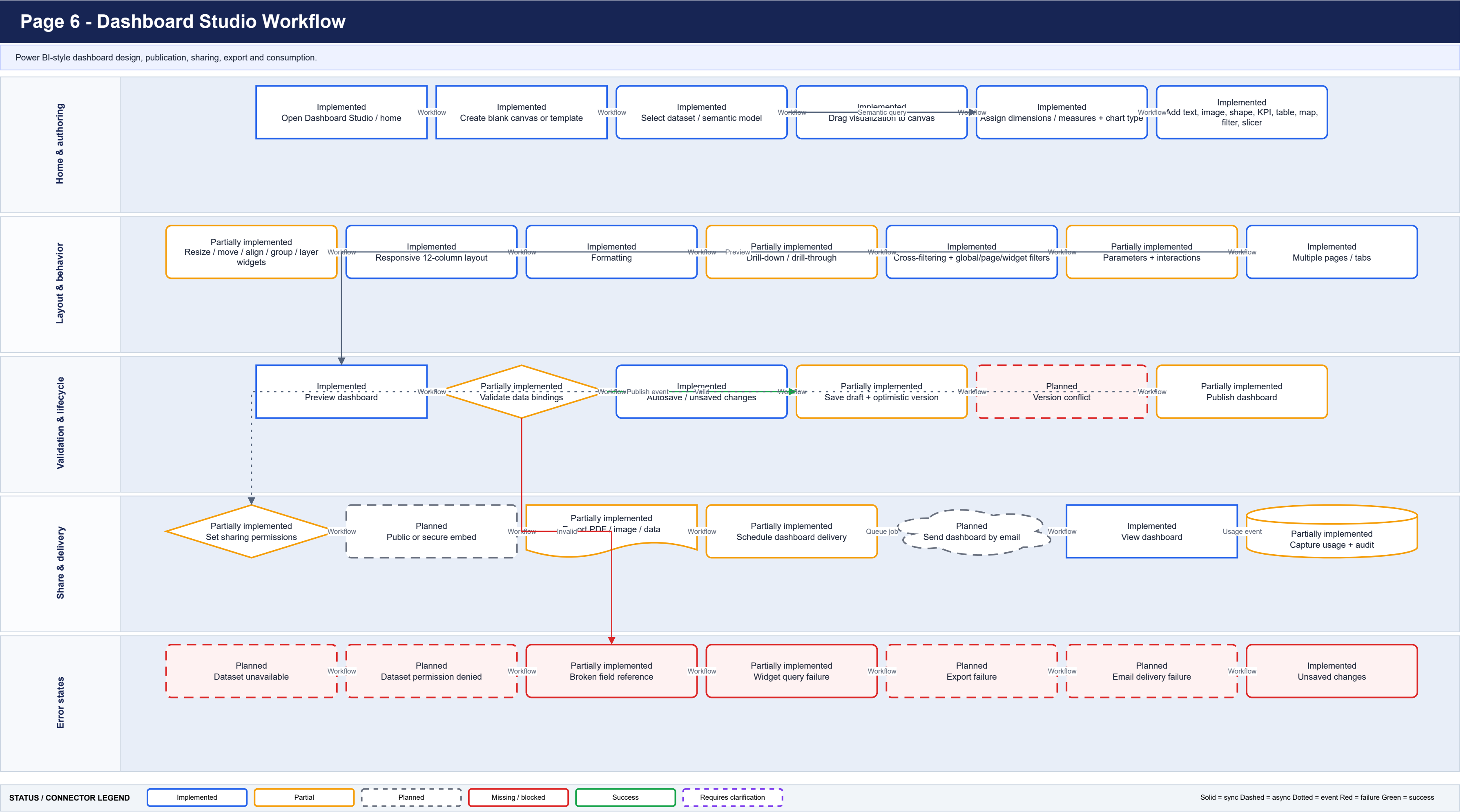
Governed dataset lifecycle and reusable business semantic layer.



STATUS / CONNECTOR LEGEND

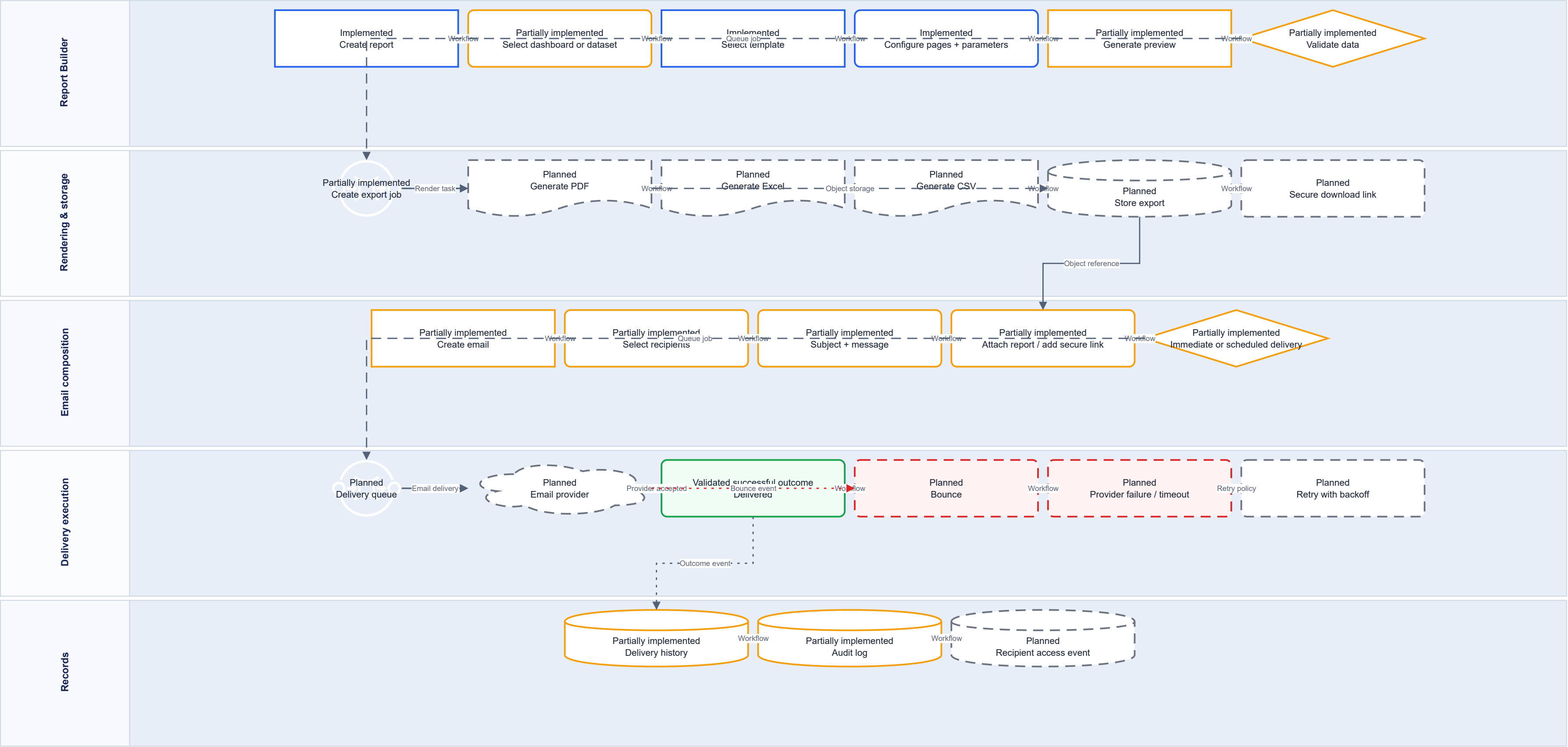


Solid = sync Dashed = async Dotted = event Red = failure Green = success



Page 7 - Report and Email Delivery Workflow

Report composition, export generation and controlled outbound delivery.



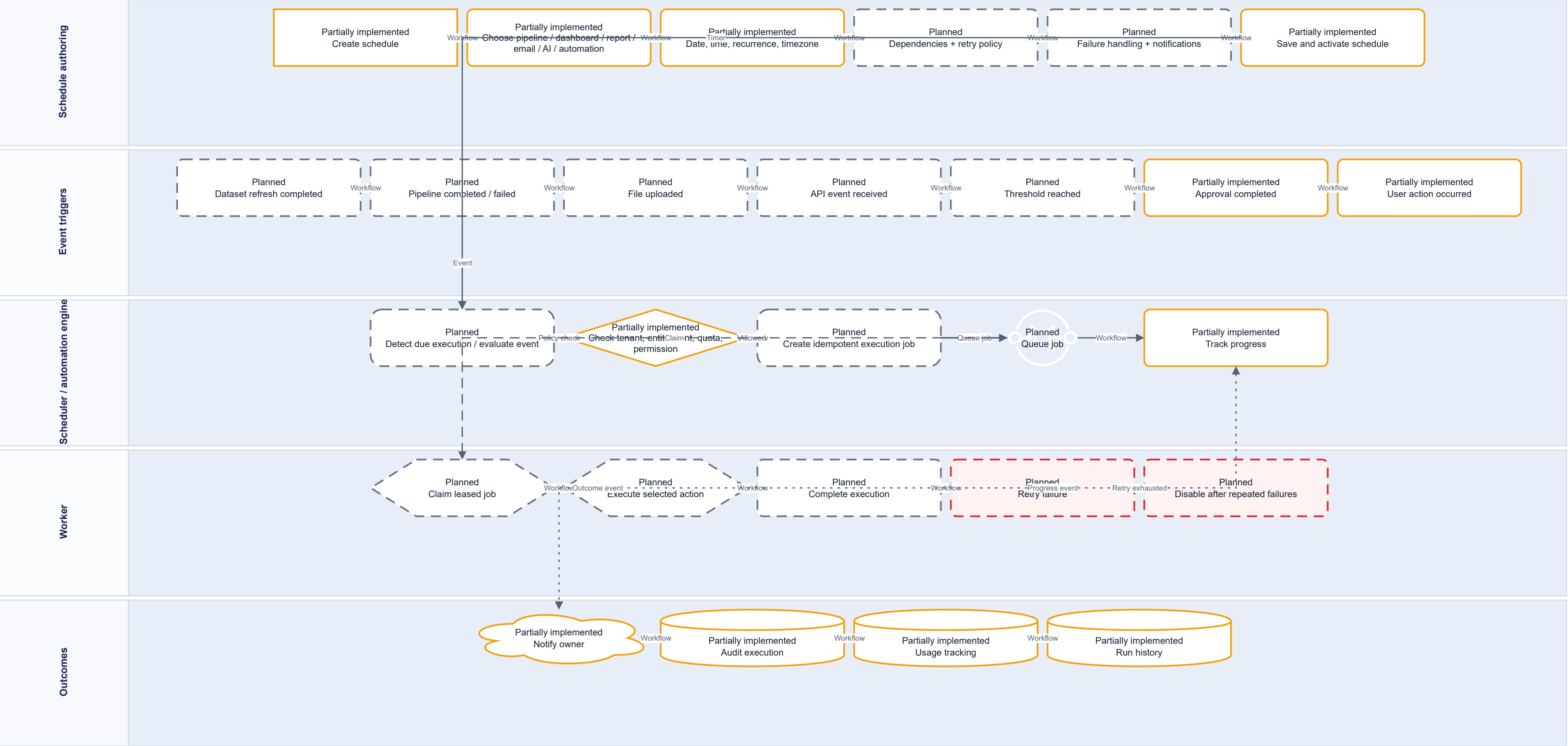
STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 8 - Scheduler and Automation Workflow

Time-based and event-driven orchestration across VIP actions.



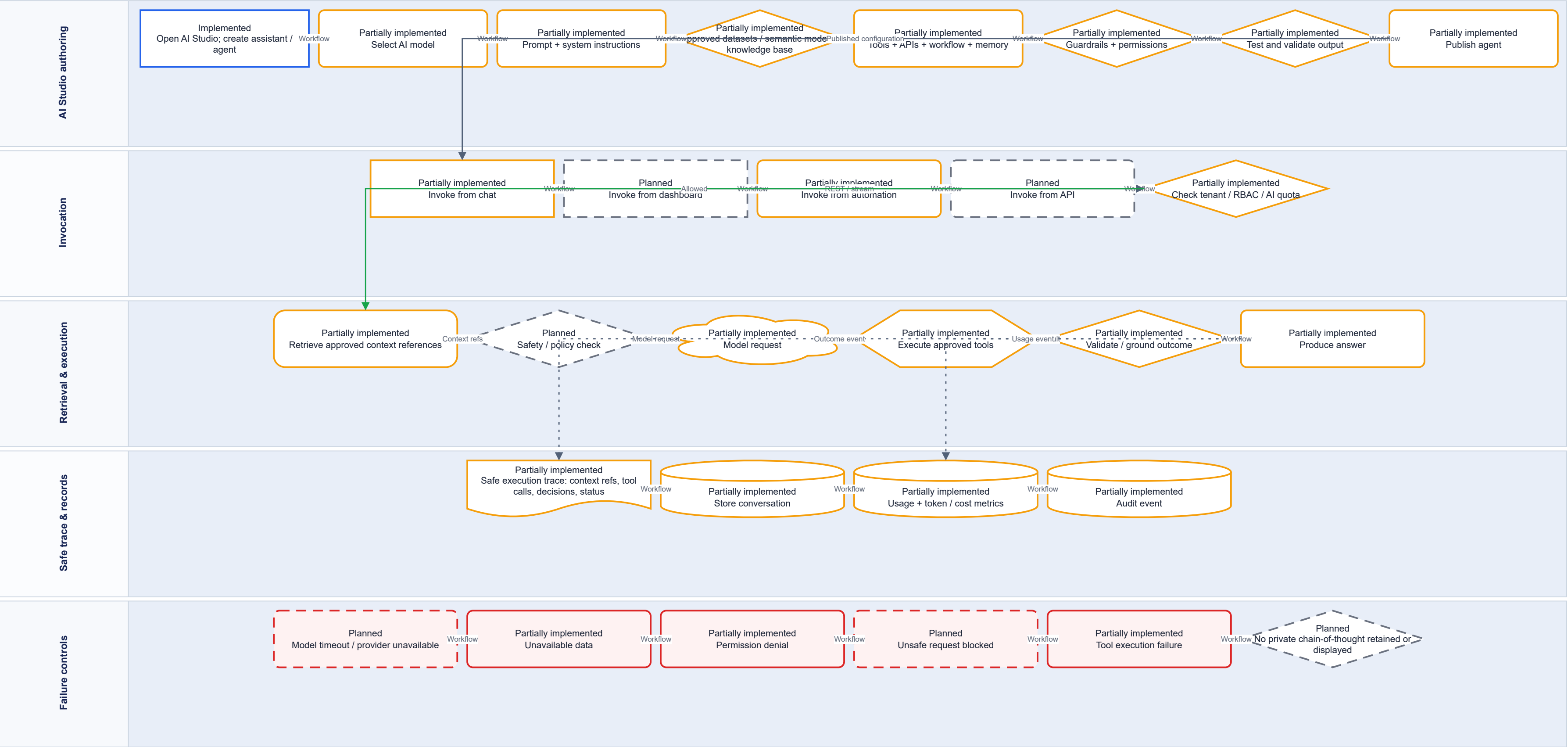
STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 9 - AI Studio and AI Agent Workflow

Permission-safe AI configuration, invocation, tool use and auditable outcomes without exposing chain-of-thought.

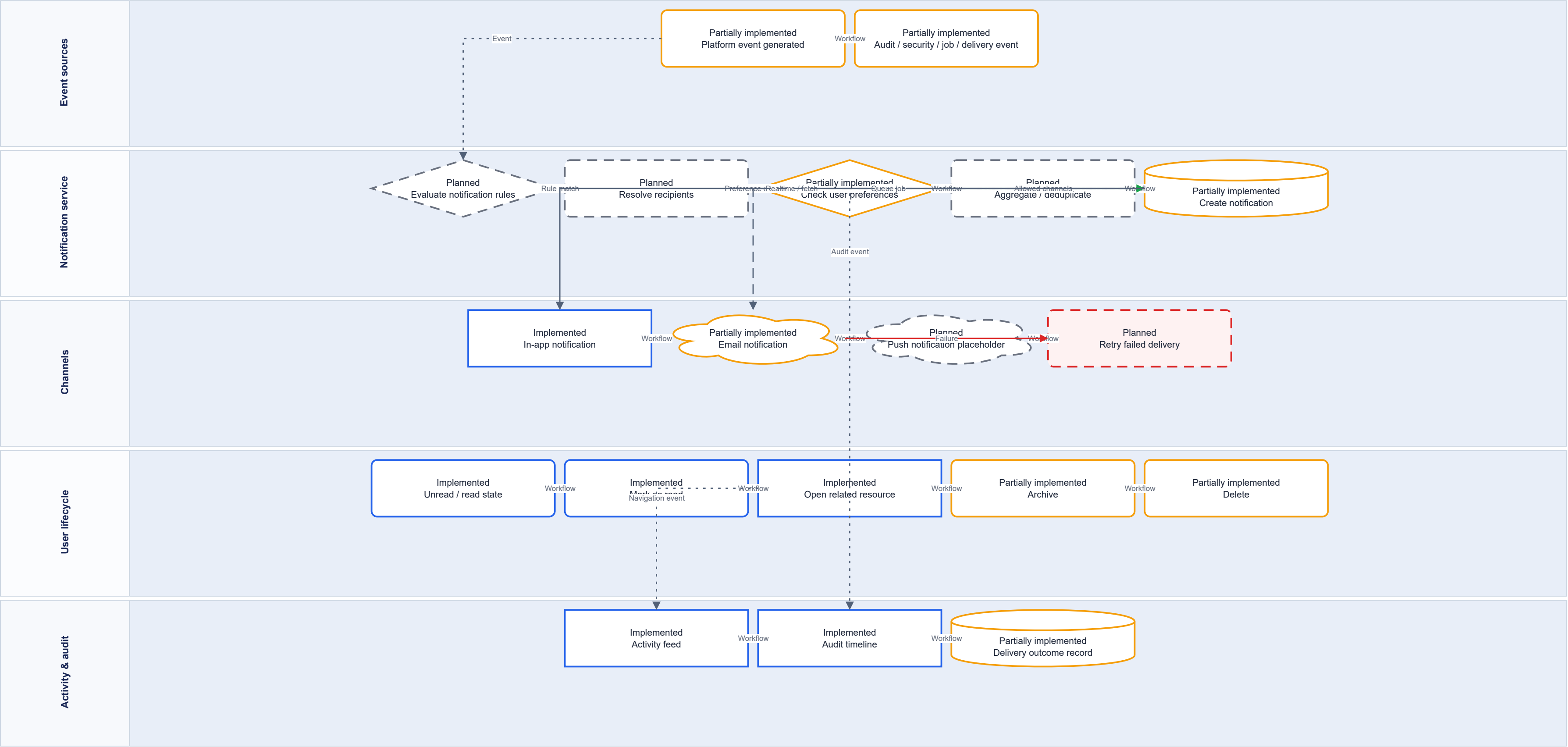


STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Rule-driven notification creation, channel delivery and user lifecycle.



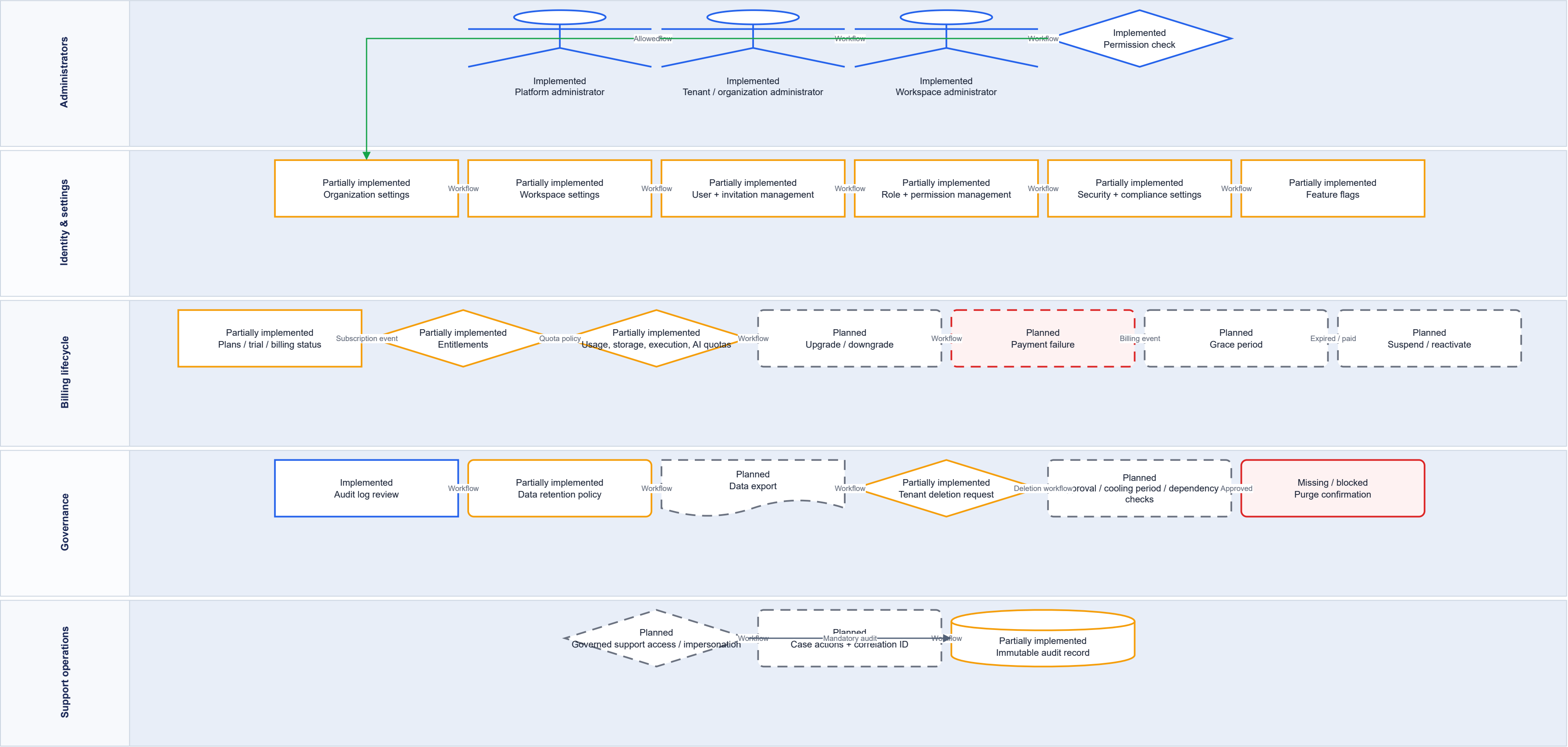
STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 11 - Administration, Billing and Governance Workflow

Tenant administration, commercial lifecycle, quota governance and controlled deletion.



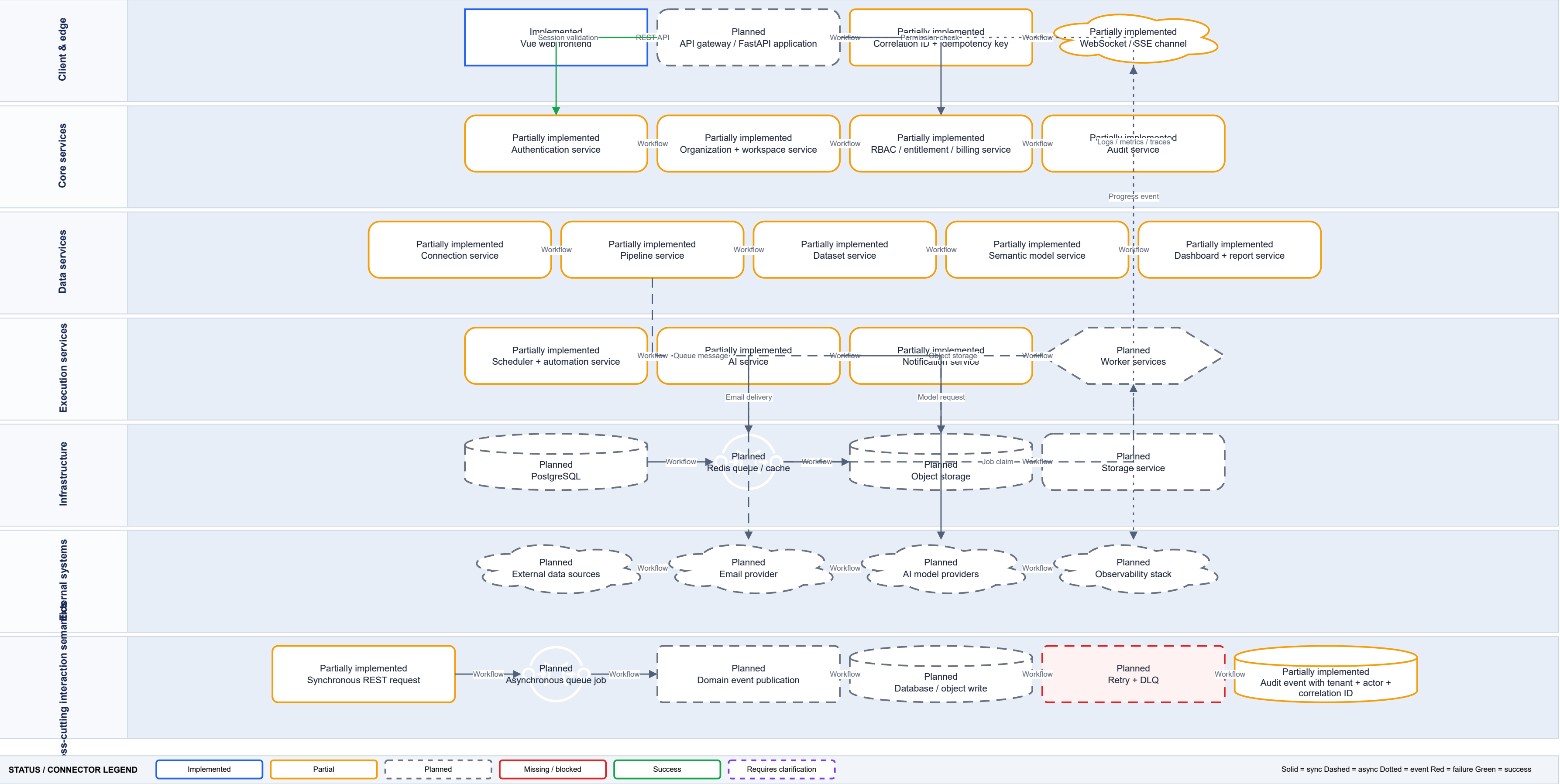
STATUS / CONNECTOR LEGEND



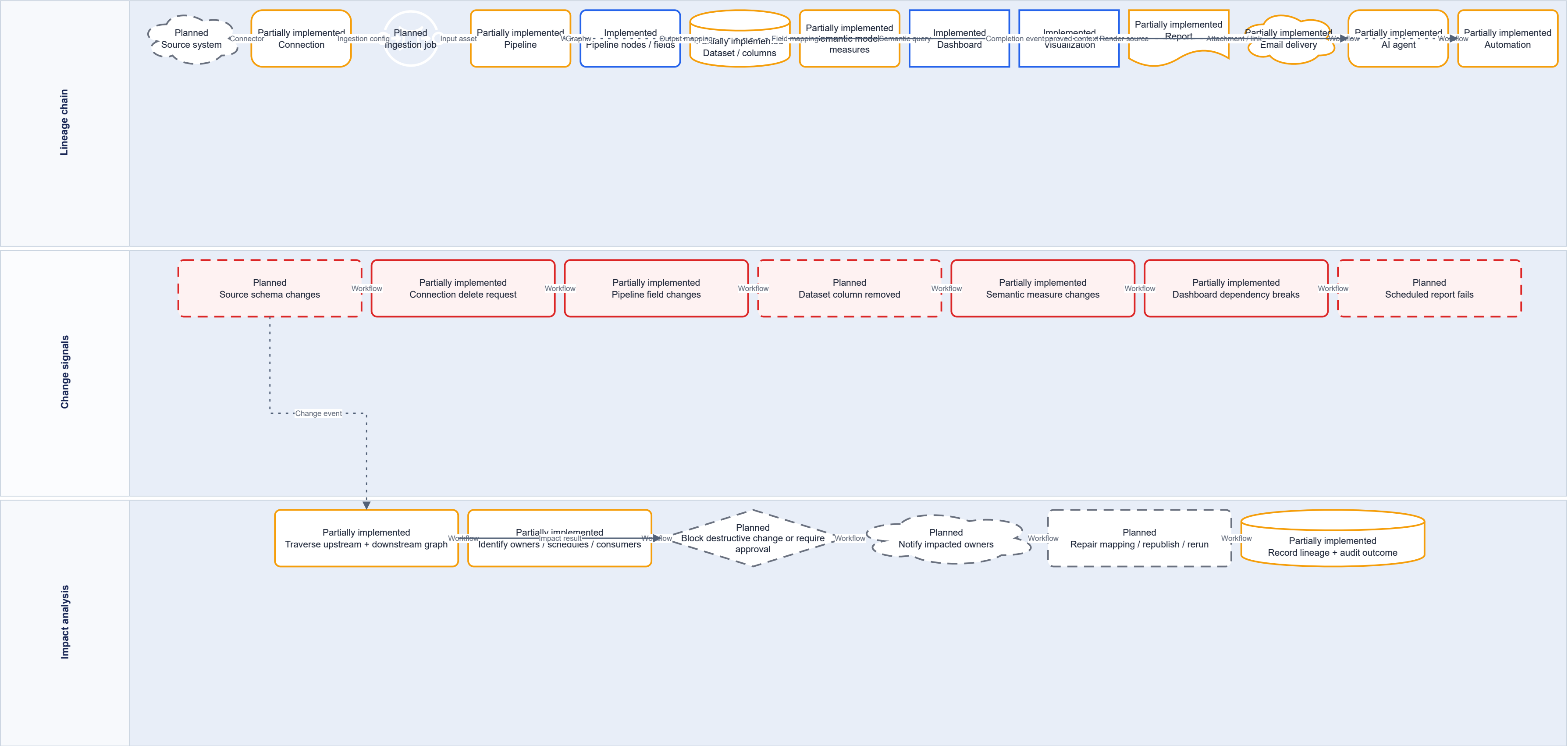
Solid = sync Dashed = async Dotted = event Red = failure Green = success

Page 12 - Backend Service Interaction Workflow

Expected target service interactions; not evidence of an implemented backend.

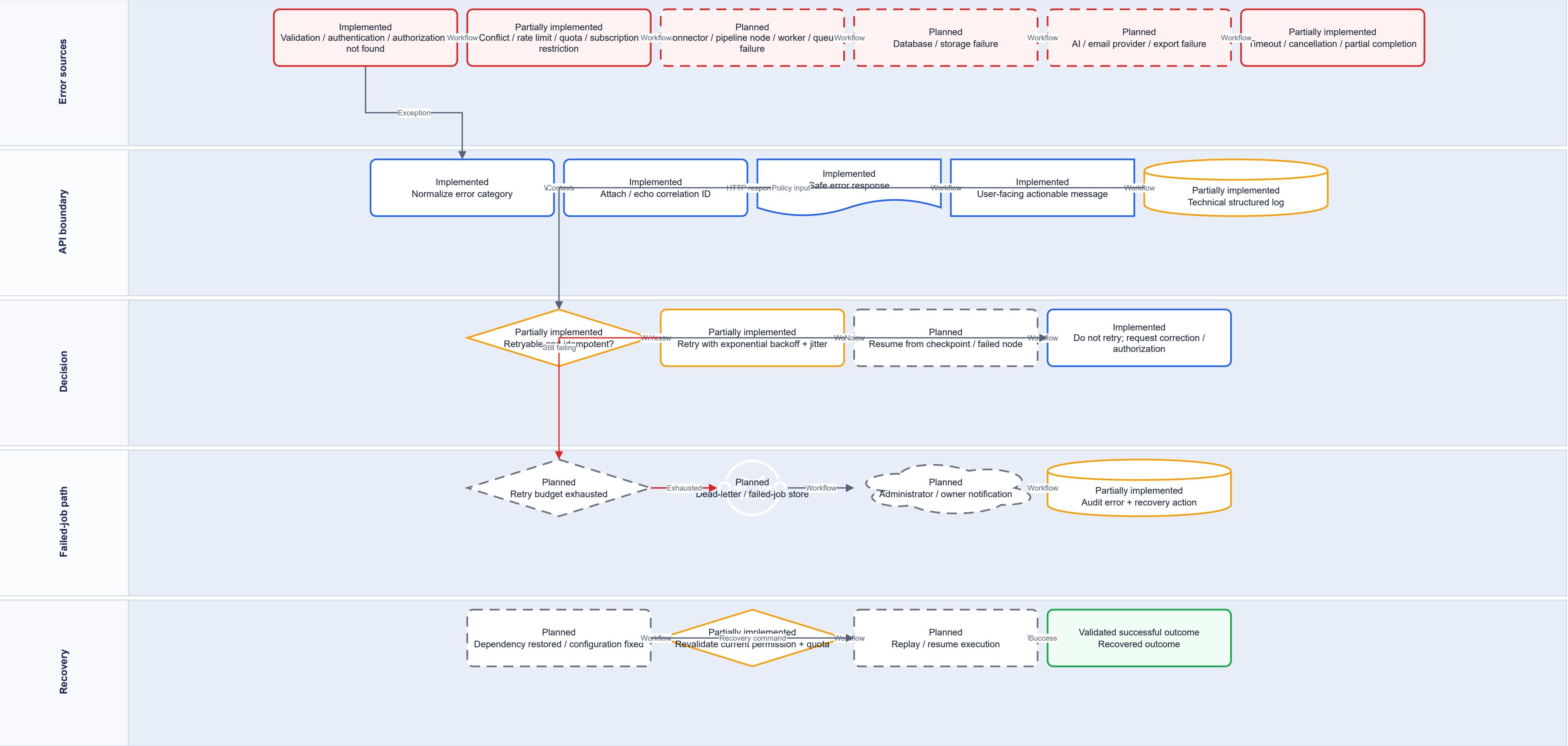


End-to-end asset lineage, break detection and impact response.



Page 14 - Error Handling and Recovery Workflow

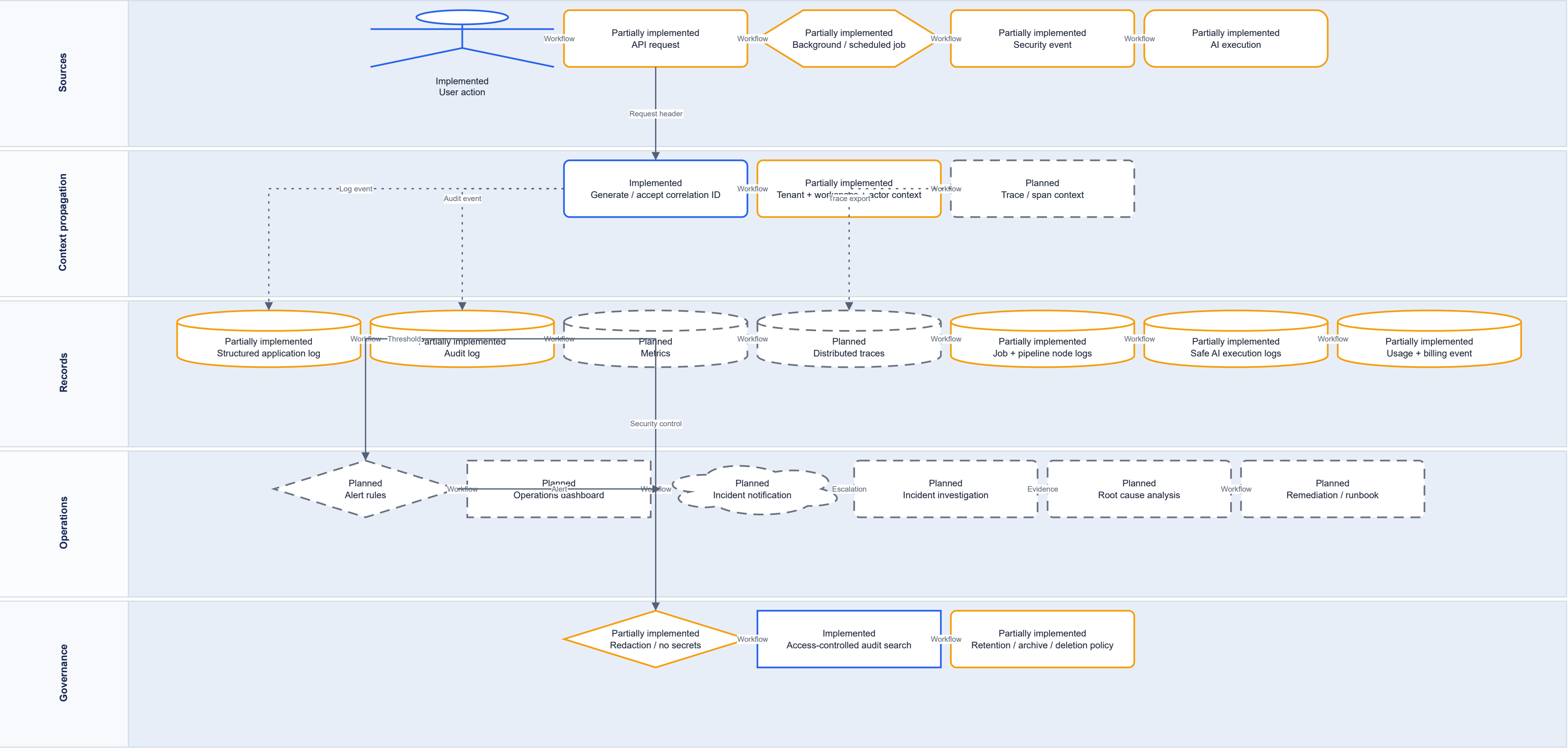
Standardized error normalization, retry, failed-job handling and recovery.



STATUS / CONNECTOR LEGEND



Evidence, telemetry and operational investigation across user and job activity.



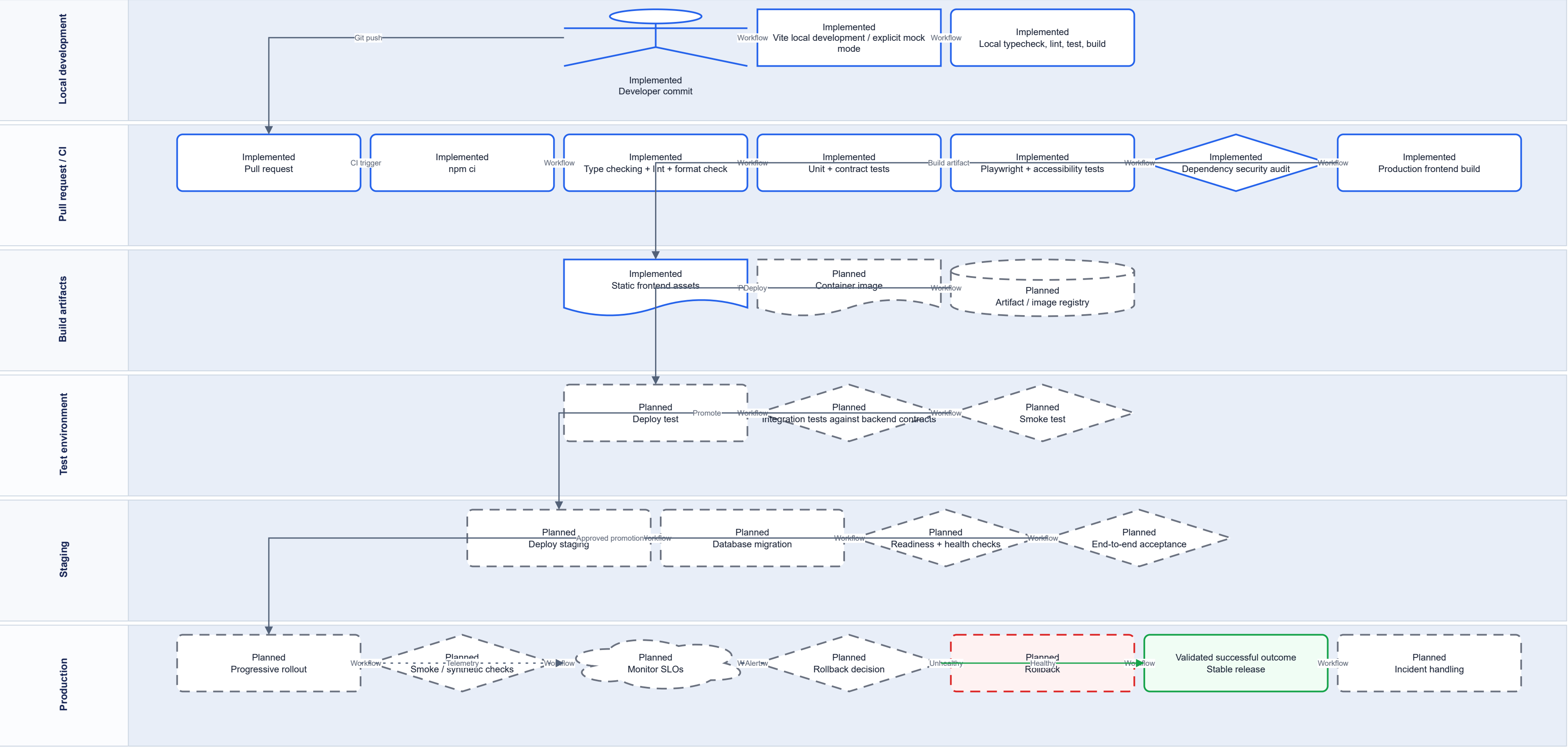
STATUS / CONNECTOR LEGEND



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Page 16 - Deployment and Runtime Workflow

Current frontend CI and target multi-environment deployment lifecycle.



STATUS / CONNECTOR LEGEND



Solid = sync Dashed = async Dotted = event Red = failure Green = success

Final Findings and Gaps

Repository comparison, risk summary and implementation priorities.

Implemented / validated frontend

- Permission-, entitlement- and feature-aware routes
- Organization/workspace switching and scoped mock persistence
- Dashboard and Pipeline Studio authoring
- Normalized errors with correlation IDs
- Unit, contract, Playwright and accessibility suites
- Frontend CI quality gate

Partial / mock-backed

- Connection, dataset, semantic, report, AI, automation, operations, admin and billing screens
- Typed service interfaces and many live REST adapter paths
- Audit, notification, usage and delivery presentation
- Publish/save lifecycles with local persistence
- Tenant context headers and session hydration

Critical production gaps

- No backend repository or proven FastAPI services
- No secure secret vault / connector execution
- No distributed queue, worker, scheduler or DLQ
- No server rendering, object storage or email delivery
- No authoritative billing/quota enforcement
- No production AI guardrail, tool sandbox or model gateway
- No deployed observability, migrations or rollout workflow

Clarifications required

- Authoritative backend language/service boundaries
- Identity provider, MFA/SSO and session policy
- Queue / worker / storage technology choices
- Billing provider and commercial state machine
- Retention, deletion, legal hold and regional residency
- AI provider, model policy, trace retention and cost allocation
- RPO/RTO, SLOs and approval ownership

PRIORITY SEQUENCE: 1) Identity + tenant authorization contract 2) Data connections + vault + ingestion 3) Queue/workers + pipeline execution 4) Dataset/semantic query service 5) Rendering/email/scheduler 6) AI governance + model gateway 7) Billing enforcement + observability + production deployment

Validation scope: XML structure checked, page inventory checked, PDF parsed and raster-rendered, visual montage inspected. See `VIP_WORKFLOW_VALIDATION_REPORT.md`.